People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research
BADJI Mokhtar University Annaba
Faculty of Medicine
General Anatomy Laboratory

Hand joints & Radiocarpal joint

Prof. A. ABDALLAH
General Anatomy Laboratory
Faculty of Medicine – Annaba
Email: abdeldjalil.abdallah@univ-annaba.dz

Course outline

- **Introduction: "The joints of the hand"**
- **Radiocarpal joint :**
  - **Introduction**
  - **Articular surfaces**
  - **Means of union**
  - **Synovial**
  - **Functional Anatomy**

Introduction: "The joints of the hand"

The joints of the hand are divided into three classes:

a. **Wrist joints :**

They include:

- Radiocarpal joint ;
- Proximal intercarpal joints ;
- Mid-carpal joint ;
- Distal intercarpal joints.

b. **Carpometacarpal joints :**

They include:

- Carpometacarpal joint of the thumb;
- Carpometacarpal joints II to V.

c. **Finger joints :**

They include:

- Metacarpophalangeal joints ;
- Interphalangeal joints.

All these joints allow for the movements of the hand and fingers: flexion, extension, adduction, abduction, circumduction.

Radiocarpal joint

1. Introduction

It is the distal joint of the thoracic limb; it unites the carpus to the forearm.

It is an ellipsoid type of synovial joints.

2. Articular surfaces

There are two of them:

a. Forearm articular surface :

It is an elliptical, concave surface with a major transverse axis. It is formed by:

- Carpal articular surface of the radius:

  It occupies the lateral 2/3 of the forearm surface, divided into two parts:

  - Lateral, triangular, joint with scaphoid bone ;
  - Medial, quadrilateral, joint with lunate bone.

- Articular disc of radioulnar joint:

  It occupies the medial 1/3 of the forearm surface.

  It is a horizontal, triangular ligament covered with cartilage. It attaches to the lower edge of the ulnar notch of the radius and to the ulnar styloid process. It lies between the ulnar head and the carpus (lunate and triquetrum).

b. Radial articular surface of the carpus:

It is convex, and more extensive than the forearm articular surface.

It is formed by the radial surfaces of the scaphoid, lunate and triquetrum bones, and by the interosseous carpal ligaments that unite them.

3. Means of union

a. Joint capsule:

It is a fibrous sleeve that envelops the joint ; it attaches to:

- The periphery of the articular surfaces;
- The anterior and posterior edges of the articular disc.

b. Passive ligaments :

- **Palmar radiocarpal ligament :**

  It attaches to the anterior edge of the carpal surface and to the radial styloid process, and terminates on the lunate and capitate.

- **Palmar ulnocarpal ligament :**

  It attaches to the articular disc and the ulnar styloid process, and terminates on the triquetrum, lunate, and capitate.

- **Dorsal radiocarpal ligament :**

  It attaches to the posterior edge of the carpal surface, and terminates on the lunate, triquetrum and hamate.

- **Radial collateral ligament of the carpus:**

  It attaches to the radial styloid process, and terminates on the scaphoid.

- **Ulnar collateral ligament of the carpus:**

  It attaches to the ulnar styloid process, and terminates on the pisiform and triquetrum.

c. Active ligaments :

Represented by the tendons of 6 main muscles:

- The radial flexor of the carpus;
- The flexor carpi ulnaris;
- The long palmaris;
- The extensor carpi ulnaris;
- The long radial extensor of the carpus;
- The short radial extensor of the carpus.

4. Synovial

It is a gliding membrane that lines the deep surface of the capsule ; it attaches to the boundary of the antebrachial and carpal articular surfaces.

Noticed: Synovial cyst

It is a rounded swelling, prominent in flexion and disappearing in extension. It is formed by synovial extensions that insinuate themselves between the capsular and ligamentous fibers.

5. Functional Anatomy

The radiocarpal joint allows for two types of movement:

a. Flexion-Extension :

They are carried out around a transverse axis.

- Flexion: Bring the palm of hand closer to the front of forearm, his amplitude is 50°.
- Extension: Bring the back of hand closer to the back of forearm, his amplitude is 35°.

b. Abduction-Adduction :

They take place around a sagittal axis.

- Abduction (or radial tilt): the hand tilts to the radial side, his amplitude is 5°.
- Adduction (or ulnar tilt): the hand tilts to the ulnar side, his amplitude is 15°.

Notice: Circumduction

It is a circular summation movement, drawing a cone. It is achieved by transitioning from one to the other of the 4 previous movements.